

relay will be activated as long as pin D12 is held high. A readymade relay module can also be used instead of the

relay and transistor. However, since relay modules are activated using a low level, suitable changes need to be made in software.

The NO terminal of the relay is connected to the UV chokes so that the UV lights are switched on when the relay is activated. An indicator LED connected to pin D13 of Arduino lights up when the UV light is on.

Using the UV steriliser

The disinfection process is dependent on the duration and strength of the UV exposure. Small items like masks and keys should be exposed for a minimum of two to three minutes while for larger items the exposure should be proportionately increased. Use a small steel rack to raise the items inside the box so that UV light can reach the lower parts of the items also. The author used a spare cake rack that came as accessory to a microwave oven.

First keep the items to be disinfected on the steel rack and close the door. The display will indicate 'Door' while the door is open. The display will change to '00:00' when the door is closed. Now you can set the time using the rotary switch. Any time from 1:00 minute to 10:00 minutes can be set. If you rotate the dial of encoder further, a beep will sound as the min and max values have been set in the software.

Now press the Start switch. The relay will switch on and the timer will start the countdown. The time remaining will appear on the display. The countdown will continue for the set time and on completion, 'End' would be displayed and a long beep sound heard. Now you may open the door and take out the items.

On closing the door, the time period will be set automatically to the last setting. If desired, it can be changed

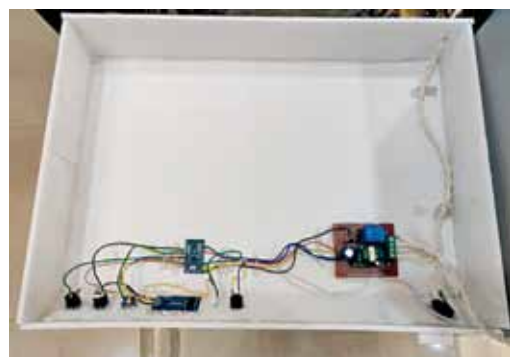


Fig. 9: The circuit inside the lid cover



Fig. 10: Sequence of message display

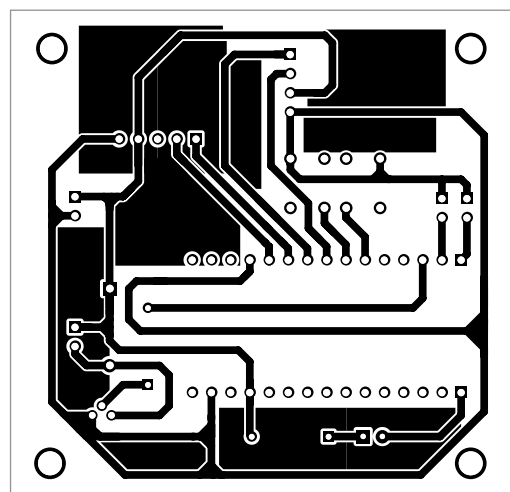


Fig. 11: Actual-size PCB layout

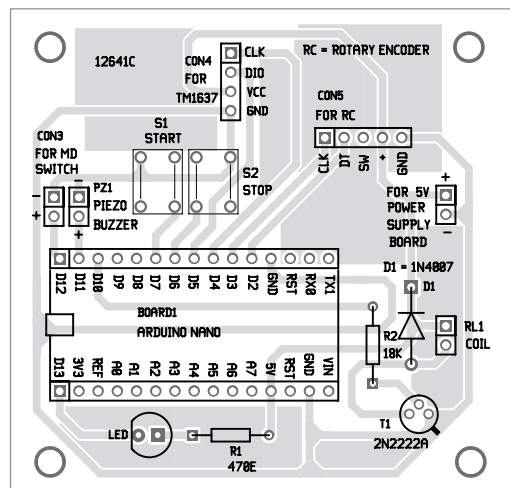


Fig. 12: Component layout of the PCB

PARTS LIST

Semiconductors:

Board1	- Arduino Nano
T1	- 2N2222A NPN transistor
D1	- 1N4007 rectifier diode
LED1	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 470-ohm
R2	- 18-kilo-ohm

Miscellaneous:

TM1637	- 4-digit display module
CON1-CON3	- 2-pin connector
CON4	- 4-pin connector
CON5	- 5-pin connector
RC	- 5-pin rotary encoder
PZ1	- Piezo buzzer
	- 5V DC power supply board
	- 230V AC operated 11W UV tube with choke

using the rotary switch, or more items may be disinfected with the old time setting. Shadows can restrict disinfection, so do not keep too many items in one go to avoid shadows.

If Stop button is pressed while the countdown is going on, the display will show 'Stop.' On pressing 'Start' button the countdown will resume and complete as usual. However, if door is opened during sterilisation, the process will be cancelled and on closing the door, the last set time will be restored. The UV lights shut off and sanitisation process stops as soon as the door is opened. While setting the time, it can be reset to '00:00' by pressing the 'Stop' button.

An actual-size PCB layout for this UV sterilisation box is shown in Fig. 11 and its component layout in Fig. 12. Before assembling the circuit do not forget to upload *UVTimer.ino* into Arduino Nano. Assemble the circuit on PCB and enclose the PCB along with Arduino and other parts in suitable UV box as explained above.

The author has been using the sanitiser to sterilise face masks after every use. Milk and other packets are also sanitised by him before use. So go ahead and sanitise your items using this simple sanitiser.

Caution. Exposure to UV-C rays is harmful to humans. Do not try to test or expose yourself to UV lights directly. **EFY**



Ashok Bajjal is a regular contributor to EFY. He currently retired and is now working on home automation projects